

Once you have the bitmap, you could make use of tess-two methods for the text extraction.

A sample code is shown below:

```
TessBaseAPI tessTwo = new TessBaseAPI();
tessTwo.init(Environment.getExternalStorageDirectory().
toString(), "eng");
tessTwo.setPageSegMode(TessBaseAPI.PageSegMode.PSM_SINGLE_
LINE);
tessTwo.setImage(bitmap);
String recognizedText = tessTwo.getUTF8Text();
tessTwo.end();
```

The TessBaseAPI instance creation shown in the first line of the above snippet will try to load some of the .so libraries of NDK and, if this fails, the app will crash without even throwing any exception, at times. So make sure the NDK is all compiled and the path is set in Android Studio before building the app.

Initialise the TessBaseAPI as shown in the second line of the code to use the path where the tessdata (tesseract trained data) is present. The tessdata is a compilation of all possible image patterns that Tesseract will match against to recognise the text. You may obtain tessdata from <https://github.com/tesseract-ocr/tessdata> for your language of interest and place it in a folder called tessdata, which is under the path that will be mentioned in the *init()* method, as in the second line above. In this article, I have used the internal storage to store the English version of tessdata, and the path to tessdata is *<path returned by Environment.getExternalStorageDirectory().toString()>/tessdata/eng.traineddata*. The *init()* method in the

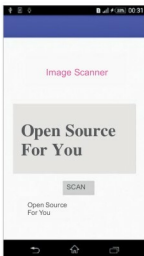


Figure 2: Sample application

gets extracted from the captured image with just the click of the camera. **END**

second line takes in 'eng' for *eng.traineddata*.

The third line sets the page segmentation mode using the method *setPageSegMode()* which tells the Tesseract engine how to treat the image — if it's a single line, a single word or a block, and so on. Finally, set the bitmap image to be processed using *setImage()* and get the text using *getUTF8Text()*.

I've made a sample app, as shown in Figure 2, where clicking the *Scan* button pops up the camera, and the text

References

- [1] [https://en.wikipedia.org/wiki/Tesseract_\(software\)](https://en.wikipedia.org/wiki/Tesseract_(software))
- [2] <https://zdenop.github.io/tesseract-doc/index.html>
- [3] <https://github.com/rmtheis/tess-two>
- [4] <https://github.com/rmtheis/tess-two/tree/master/tess-two-test/src/com/googlecode/tesseract/android/test>

By: Shravan I.V.

The author currently works as a software developer at Cisco Systems India. He is interested in open source technologies and can be reached at iv.shravan@gmail.com.

Read more stories on Components in
www.electronicb2b.com

TOP COMPONENTS STORIES

- The latest in power converters
- India's leading component distributors
- Growth of Indian electronics components industry
- The latest launches of components for LEDs
- The latest launches of components for electronics

**ELECTRONICS
B2B
INDUSTRY IS AT A**

CLICK OF A BUTTON

Log on to www.electronicb2b.com and be in touch with the Electronics B2B Fraternity 24x7